Exercise 9: Distributions and the Analytics Pane

Three worksheets, three captions. No memo.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 9 (trends, clustering and distributions). Tableau Help: Bin a continuous measure; Box plots; Trend lines and Describe Trend Model.
Starter workbook	MIS6380_Exercise9_Starter.twbx: data connected, student_id set as a Dimension, Graduation Rate and Part-Time Share ready to use.
Walkthrough	Module 9 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a caption on each. No memo.

Before you start

Every exercise so far has compared averages: graduation rates, aid per credit, withdrawal by campus. An average is one number standing in for hundreds of students. This exercise looks behind it: how values are spread out, whether two groups really differ, and how much a trend line explains. All three tools live in the Analytics pane or one right-click away.

Set up

1. Open the Exercise 9 starter workbook, or connect mvu_student_dim.csv and set student_id to a Dimension.
2. Check (or create) **Graduation Rate** = AVG([is_graduated]) and **Part-Time Share** = AVG(IIF([enrollment_intensity] = "Part-Time", 1, 0)), both formatted as Percentage, 1 decimal.

Checkpoint: How many students have total_aid_awarded of exactly $0?

Worksheet 1: a histogram, and how bin size changes it

1. Create an Integer parameter **Bin Size**: current value 5000, allowable values a List of 2500, 5000 and 10000. Show the parameter control.
2. Right-click total_aid_awarded > Create > Bins. Name it **Aid (bin)** and set Size of bins to the Bin Size parameter.
3. Sheet **1 Histogram**: Aid (bin) on Columns, student_id as Count (Distinct) on Rows, labels on. Insert the parameter in the title: “Students by total aid, bins of $<Parameters.Bin Size>”.

Checkpoint: With Bin Size 5,000, how many students are in the first bin ($0 to $4,999)?

4. Switch Bin Size to 2,500, then to 10,000. Watch the first bar each time.

Checkpoint: With Bin Size 2,500, how many students are in the first bin, and how many in the second?

5. Create **Has Aid Record** = [total_aid_awarded] > 0. Sheet **1 Summary**: Has Aid Record on Columns; Measure Values on Text, showing Count (Distinct) of student_id, Average of total_aid_awarded and Median of total_aid_awarded. Turn on Analysis > Totals > Show Column Grand Totals.

Checkpoint: Average aid for students with an aid record, and for all students?

6. Dashboard **Worksheet 1**: the histogram, the summary and the Bin Size control.

Worksheet 2: box plots, and whether groups really differ

1. Sheet **2 Box Plot**: college on Columns, total_aid_awarded (Sum) on Rows, student_id on Detail, so there is one mark per student. Mark type Circle, opacity about 40%.
2. Analytics pane: drag **Box Plot** onto Cell. Keep the whiskers at data within 1.5 times the IQR.
3. Hover over each box to read its median, quartiles and whiskers. Hover over the Unassigned box and count its marks.

Checkpoint: What is the median aid in the College of Education?

4. Add enrollment_intensity to Columns, after college, and color the circles by it.

Checkpoint: In how many of the five colleges is the part-time median higher than the full-time median?

Worksheet 3: trend lines, and what R^2 and the p-value mean

1. Sheet **3A Part-Time**: Part-Time Share on Columns, Graduation Rate on Rows, program_name on Detail, so there is one mark per program. Filter cohort_year to 2020 and 2021, and exclude Unknown Program (4 students).

Checkpoint: How many marks does the sheet draw?

2. Analytics pane: drag **Trend Line > Linear** onto the view. Right-click the line > **Describe Trend Model** and find R-Squared and the model p-value.

Checkpoint: R-Squared and p-value for the 2020–21 cohorts?

3. Remove the cohort filter for a moment and read the trend model again. Then put the filter back.
4. Duplicate the sheet as **3B GPA** and replace Part-Time Share with AVG(undergrad_gpa). Read its trend model.

Checkpoint: R-Squared and p-value for average GPA vs. graduation rate, 2020–21?

Your three captions

Put each caption in a Text object on the dashboard, or in the sheet caption (Worksheet > Show Caption): one or two sentences answering each question.

Sheet	What your caption must answer
Worksheet 1	What does the first bar show at each bin size, and which size would you publish? Why do the mean and median differ, and should the $0 students be in the average?
Worksheet 2	Do the colleges really differ in aid? Compare the gap between their medians with the height of the boxes. What changes when you split by enrollment intensity?
Worksheet 3	What does the part-time trend line tell a dean, and what can't it tell them?

Sheet	What your caption must answer
	Why do R^2 and the p-value weaken when all cohorts are included, and why is GPA a weaker story?

Worksheet 3's caption carries 25% on its own. Reporting R^2 and the p-value without saying what they can't show does not earn full marks. Say what each mark is (a program, not a student), why a trend isn't a cause, and how the 2022 cohort dilutes it.

Before you submit: confirm every caption is visible.

Submit for Exercise 9

- Sheet names: 1 Histogram · 1 Summary · 2 Box Plot · 3A Part-Time · 3B GPA, plus the Worksheet 1 dashboard
- Worksheet 1: a histogram driven by the Bin Size parameter, with a mean-and-median summary
- Worksheet 2: a box plot of aid by college, split by enrollment intensity
- Worksheet 3: two program-level scatters with linear trend lines, 2020–21 cohorts, Unknown Program excluded
- A visible caption on each
- Packaged workbook (.twbx). No memo.

Optional extensions

- Analytics pane > **Cluster** on sheet 3A. How many clusters does Tableau suggest, and do they mean anything a dean could act on?
- Add a reference distribution (60th and 80th percentiles) to the box plot, and say what it adds.